

# **Commerce XI Maths App**

## **Theory (Pages 10–15)**

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 1 - Sets and Relations | Total questions: 35

## Question 1 - Theory Q1

In an ordered pair  $(a, b)$ , why is the order important?

Options

- A.  $(a, b)$  and  $(b, a)$  are generally different
- B. The components must be equal
- C. Only the second component is used
- D. The components must be natural numbers

**Correct answer:  $(a, b)$  and  $(b, a)$  are generally different**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives  $(a, b)$  and  $(b, a)$  are generally different. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 2 - Theory Q2

In the ordered pair  $(a, b)$ ,  $a$  is called the:

Options

- A. First component
- B. Second component
- C. Image only
- D. Co-domain

**Correct answer: First component**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives First component. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 3 - Theory Q3

Two ordered pairs  $(a, b)$  and  $(c, d)$  are equal if and only if:

Options

- A.  $a = c$  and  $b = d$
- B.  $a = d$  and  $b = c$
- C.  $a = b$  and  $c = d$
- D.  $a + b = c + d$

**Correct answer:  $a = c$  and  $b = d$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives  $a = c$  and  $b = d$ . The other choices omit a required element, include an invalid element, or use a different operation.

## Question 4 - Theory Q4

$(a, b) = (b, a)$  if and only if:

Options

A.  $a = b$

B.  $a \neq b$

C.  $a = 0$

D.  $b = 0$

**Correct answer:  $a = b$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives  $a = b$ . The other choices omit a required element, include an invalid element, or use a different operation.

## Question 5 - Theory Q5

The Cartesian product  $A \times B$  is the set of:

Options

A. All ordered pairs  $(a, b)$  where  $a \in A$  and  $b \in B$

B. All unordered pairs from A and B

C. Elements common to A and B

D. All subsets of  $A \cup B$

**Correct answer: All ordered pairs  $(a, b)$  where  $a \in A$  and  $b \in B$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives All ordered pairs  $(a, b)$  where  $a \in A$  and  $b \in B$ . The other choices omit a required element, include an invalid element, or use a different operation.

## Question 6 - Theory Q6

How is the Cartesian product of A and B read?

Options

- A. A cross B
- B. A union B
- C. A intersection B
- D. A related to B

**Correct answer: A cross B**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives A cross B. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 7 - Theory Q7

If  $A = \{1, 2\}$  and  $B = \{a, b\}$ , which ordered pair belongs to  $A \times B$ ?

Options

- A. (2, b)
- B. (a, 1)
- C. (b, 2)
- D. (a, b)

**Correct answer: (2, b)**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives (2, b). The other choices omit a required element, include an invalid element, or use a different operation.

## Question 8 - Theory Q8

If  $A = \emptyset$  or  $B = \emptyset$ , then  $A \times B$  is:

Options

- A.  $\emptyset$
- B. A
- C. B
- D.  $\{\emptyset\}$

**Correct answer:**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives  $\emptyset$ . The other choices omit a required element, include an invalid element, or use a different operation.

## Question 9 - Theory Q9

For finite sets A and B, which formula is correct?

Options

- A.  $n(A \times B) = n(A) \times n(B)$
- B.  $n(A \times B) = n(A) + n(B)$
- C.  $n(A \times B) = n(A) - n(B)$
- D.  $n(A \times B) = 2^{n(A)}$

**Correct answer:  $n(A \times B) = n(A) \times n(B)$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives  $n(A \times B) = n(A) \times n(B)$ . The other choices omit a required element, include an invalid element, or use a different operation.

## Question 10 - Theory Q10

A relation R from a non-empty set A to a non-empty set B is:

Options

- A. Any subset of  $A \times B$
- B. Always equal to  $A \times B$
- C. Any subset of  $A \cup B$
- D. A subset of  $B \times A$  only

**Correct answer: Any subset of  $A \times B$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Any subset of  $A \times B$ . The other choices omit a required element, include an invalid element, or use a different operation.

## Question 11 - Theory Q11

If  $(x, y) \in R$ , the notation  $xRy$  is read as:

Options

- A. x is related to y under R
- B. x is equal to y
- C. x is a subset of y
- D. x is the image of R

**Correct answer: x is related to y under R**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives x is related to y under R. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 12 - Theory Q12

If  $(x, y) \in R$ , then y is called the:

Options

- A. Image of x under R
- B. Pre-image of x
- C. Domain of R
- D. First component of x

**Correct answer: Image of x under R**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Image of x under R. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 13 - Theory Q13

If  $(x, y) \in R$ , then x is called the:

Options

- A. Pre-image of y under R
- B. Image of y
- C. Range of R
- D. Second component of y

**Correct answer: Pre-image of y under R**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Pre-image of y under R. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 14 - Theory Q14

The domain of a relation R is the set of all:

Options

- A. First components of ordered pairs in R
- B. Second components of ordered pairs in R
- C. Elements in the co-domain
- D. Subsets of R

**Correct answer: First components of ordered pairs in R**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives First components of ordered pairs in R. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 15 - Theory Q15

If R is a relation from A to B, its co-domain is:

Options

- A. B
- B. A
- C. R
- D.  $A \times B$

**Correct answer: B**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives B. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 16 - Theory Q16

The range of a relation R is the set of all:

Options

- A. Second components of ordered pairs in R
- B. First components of ordered pairs in R
- C. Elements of  $A \times B$  not in R
- D. Subsets of the domain

**Correct answer: Second components of ordered pairs in R**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Second components of ordered pairs in R. The other choices omit a required element, include an invalid

element, or use a different operation.

## Question 17 - Theory Q17

A relation is one-one when:

Options

- A. Distinct elements in the domain have distinct images
- B. Every element has the same image
- C. The range is a proper subset of the co-domain
- D. No element has an image

**Correct answer: Distinct elements in the domain have distinct images**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Distinct elements in the domain have distinct images. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 18 - Theory Q18

A relation is many-one when:

Options

- A. Two or more domain elements have the same image
- B. Every domain element has a distinct image
- C. Every co-domain element has a pre-image
- D. The domain and range are equal

**Correct answer: Two or more domain elements have the same image**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Two or more domain elements have the same image. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 19 - Theory Q19

A relation from A to B is into when:

Options

- A. Its range is a proper subset of B
- B. Its range is equal to B
- C. Its domain is empty

D. It is equal to  $A \times B$

**Correct answer: Its range is a proper subset of B**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Its range is a proper subset of B. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 20 - Theory Q20

A relation from A to B is onto when:

Options

- A. Every element of B is the image of some element of A
- B. Every element of A has two images
- C. Its range is a proper subset of B
- D. Its domain and co-domain are disjoint

**Correct answer: Every element of B is the image of some element of A**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Every element of B is the image of some element of A. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 21 - Theory Q21

A binary relation on a non-empty set A is:

Options

- A. Any subset of  $A \times A$
- B. Any subset of A only
- C. Only the empty set
- D. Any subset of  $A \times B$  where  $B \neq A$

**Correct answer: Any subset of  $A \times A$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Any subset of  $A \times A$ . The other choices omit a required element, include an invalid element, or use a different operation.

## Question 22 - Theory Q22

Which is the universal relation on A?

Options

A.  $A \times A$

B.  $\emptyset$

C.  $A \cup A$

D.  $P(A)$

**Correct answer:  $A \times A$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives  $A \times A$ . The other choices omit a required element, include an invalid element, or use a different operation.

### Question 23 - Theory Q23

If  $n(A) = m$  and  $n(B) = n$ , the number of relations from A to B is:

Options

A.  $2^{mn}$

B.  $m \times n$

C.  $2^m + 2^n$

D.  $m^n$

**Correct answer:  $2^{mn}$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives  $2^{mn}$ . The other choices omit a required element, include an invalid element, or use a different operation.

### Question 24 - Theory Q24

A relation R on A is reflexive when:

Options

A.  $(a, a) \in R$  for every  $a \in A$

B.  $(a, b) \in R$  implies  $(b, a) \in R$

C.  $(a, b) \in R$  and  $(b, c) \in R$  imply  $(a, c) \in R$

D. No pair  $(a, a)$  belongs to R

**Correct answer:  $(a, a) \in R$  for every  $a \in A$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives  $(a, a) \in R$  for every  $a \in A$ . The other choices omit a required element, include an invalid element, or use a different operation.

## Question 25 - Theory Q25

A relation  $R$  on  $A$  is symmetric when:

Options

- A.  $(a, b) \in R$  implies  $(b, a) \in R$
- B.  $(a, a) \in R$  for every  $a \in A$
- C.  $(a, b) \in R$  implies  $a = b$
- D.  $(a, b) \in R$  and  $(b, c) \in R$  imply  $(a, c) \in R$

**Correct answer:  $(a, b) \in R$  implies  $(b, a) \in R$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives  $(a, b) \in R$  implies  $(b, a) \in R$ . The other choices omit a required element, include an invalid element, or use a different operation.

## Question 26 - Theory Q26

A relation  $R$  on  $A$  is transitive when:

Options

- A.  $(a, b) \in R$  and  $(b, c) \in R$  imply  $(a, c) \in R$
- B.  $(a, b) \in R$  implies  $(b, a) \in R$
- C.  $(a, a) \in R$  for every  $a \in A$
- D. Every element has exactly one image

**Correct answer:  $(a, b) \in R$  and  $(b, c) \in R$  imply  $(a, c) \in R$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives  $(a, b) \in R$  and  $(b, c) \in R$  imply  $(a, c) \in R$ . The other choices omit a required element, include an invalid element, or use a different operation.

## Question 27 - Theory Q27

An equivalence relation is:

Options

- A. Reflexive, symmetric, and transitive

- B. Only reflexive and symmetric
- C. Only symmetric and transitive
- D. One-one and onto

**Correct answer: Reflexive, symmetric, and transitive**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Reflexive, symmetric, and transitive. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 28 - Theory Q28

For arbitrary non-empty sets A and B, which statement is generally true?

Options

- A.  $A \times B \neq B \times A$
- B.  $A \times B = B \times A$  always
- C.  $A \times B = A \cap B$
- D.  $A \times B = A \cup B$

**Correct answer:  $A \times B \neq B \times A$**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives  $A \times B \neq B \times A$ . The other choices omit a required element, include an invalid element, or use a different operation.

## Question 29 - Theory Q29

If  $n(A) = 4$  and  $n(B) = 3$ , enter  $n(A \times B)$ .

**Correct answer: 12**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 12. The other choices omit a required element, include an invalid element, or use a different operation.

## Question 30 - Theory Q30

If  $n(A) = 2$  and  $n(B) = 3$ , enter the total number of relations from A to B.

**Correct answer: 64**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 64. The other choices omit a required element, include an invalid element, or use a different operation.

### Question 31 - Theory Q31

For  $A = \{1, 2, 3, 4, 5, 6\}$  and  $R = \{(x, y) / y = x + 1\}$ , enter  $n(R)$ .

**Correct answer: 5**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 5. The other choices omit a required element, include an invalid element, or use a different operation.

### Question 32 - Theory Q32

If  $(x + 3, 2) = (4, y - 3)$ , enter  $x$ .

**Correct answer: 1**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 1. The other choices omit a required element, include an invalid element, or use a different operation.

### Question 33 - Theory Q33

If  $(x + 3, 2) = (4, y - 3)$ , enter  $y$ .

**Correct answer: 5**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 5. The other choices omit a required element, include an invalid element, or use a different operation.

### Question 34 - Theory Q34

For  $R = \{(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}$ , enter  $n(\text{Domain}(R))$ .

**Correct answer: 4**

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 4. The other choices omit a required element, include an invalid element, or use a different operation.

### Question 35 - Theory Q35

For  $R = \{(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}$ , enter  $n(\text{Range}(R))$ .

**Correct answer: 2**

### Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives

2. The other choices omit a required element, include an invalid element, or use a different operation.